

LAZARD LEGACY MATH ACADEMY

GED Math Essentials

Focus on the Hardest GED Topics — 30 Minutes

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"Dreams can come true. Better late than never."

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Section 1: Quick Review — Fractions & Percentages

Fraction Rules — The Essentials:

Adding/Subtracting: You MUST have a common denominator first.

- Find the LCD (Least Common Denominator)
- Convert each fraction, then add/subtract the numerators

Multiplying: Multiply straight across. Numerator $\times$ Numerator, Denominator $\times$ Denominator.

Dividing: Keep the first fraction, flip the second (reciprocal), then multiply.

Percentage Conversions:

- To convert % to decimal: divide by 100 (move decimal 2 places left). Example: 25% = 0.25
- To convert decimal to %: multiply by 100 (move decimal 2 places right). Example: 0.75 = 75%
- To find % of a number: multiply the decimal form $\times$ the number. Example: 30% of 50 = $0.30 \times 50 = 15$

Worked Examples:

Example 1:

Add: $\frac{2}{3} + \frac{1}{4}$

Solution: LCD = 12. Convert: $\frac{2}{3} = \frac{8}{12}$ and $\frac{1}{4} = \frac{3}{12}$. Add: $\frac{8}{12} + \frac{3}{12} = \frac{11}{12}$.

Example 2:

Multiply: $\frac{3}{5} \times \frac{2}{7}$

Solution: Multiply straight across: $(3 \times 2) / (5 \times 7) = \mathbf{6/35}$.

Example 3:

Divide: $\frac{4}{5} \div \frac{2}{3}$

Solution: Keep, flip, multiply: $\frac{4}{5} \times \frac{3}{2} = \frac{12}{10} = \mathbf{\frac{6}{5} = 1 \frac{1}{5}}$.

Example 4:

What is 35% of 200?

Solution: Convert 35% to decimal: 0.35. Multiply: $0.35 \times 200 = \mathbf{70}$.

Example 5:

A shirt was \$60 and is now \$45. What is the percent decrease?

Solution: Change = $\$60 - \$45 = \$15$. Percent = $(\text{change/original}) \times 100 = (\frac{15}{60}) \times 100 = \mathbf{25\% \text{ decrease}}$.

Section 2: Quick Review — Algebra Equations

Solving One-Step and Two-Step Equations:

Golden Rule: Whatever you do to one side, you **MUST** do to the other side.

Steps for solving:

1. Simplify each side (distribute, combine like terms).
2. Move variable terms to one side (add/subtract).
3. Move constant terms to the other side.
4. Divide by the coefficient of the variable.
5. Check your answer by substituting back.

Worked Examples:

Example 1:

Solve: $5x + 3 = 28$

Solution: Subtract 3: $5x = 25$. Divide by 5: $x = 5$. Check: $5(5)+3 = 28$ ✓

Example 2:

Solve: $2(x - 4) = 10$

Solution: Distribute: $2x - 8 = 10$. Add 8: $2x = 18$. Divide by 2: $x = 9$. Check:
 $2(9-4) = 2(5) = 10$ ✓

Example 3:

Solve: $3x + 7 = x + 15$

Solution: Subtract x : $2x + 7 = 15$. Subtract 7: $2x = 8$. Divide by 2: $x = 4$. Check:
 $3(4)+7 = 19$, and $4+15 = 19$ ✓... wait, let me redo: $3(4)+7=19$, $4+15=19$ ✓

Example 4:

Solve: $x/3 - 2 = 5$

Solution: Add 2: $x/3 = 7$. Multiply by 3: $x = \mathbf{21}$. Check: $21/3 - 2 = 7 - 2 = 5 \checkmark$

Example 5:

Solve: $4(x + 1) - 2x = 12$

Solution: Distribute: $4x + 4 - 2x = 12$. Combine: $2x + 4 = 12$. Subtract 4: $2x = 8$.
 $x = \mathbf{4}$.

Section 3: Quick Review — Geometry Basics for the GED

GED Formula Sheet Highlights (these ARE provided on the test):

- Area of rectangle = lw
- Area of triangle = $\frac{1}{2}bh$
- Area of circle = πr^2
- Circumference = $2\pi r$
- Volume of rectangular prism = lwh
- Volume of cylinder = πr^2h
- Pythagorean theorem: $a^2 + b^2 = c^2$

Key tip: Even though formulas are given, you must know HOW to use them. Practice identifying which values plug in where.

Worked Examples:

Example 1:

Find the volume of a rectangular box: length=5, width=3, height=4.

Solution: $V = l \times w \times h = 5 \times 3 \times 4 = \mathbf{60 \text{ cubic units.}}$

Example 2:

Find the volume of a cylinder with radius 3 and height 8. ($\pi \approx 3.14$)

Solution: $V = \pi r^2h = 3.14 \times (3)^2 \times 8 = 3.14 \times 9 \times 8 = \mathbf{226.08 \text{ cubic units.}}$

Example 3:

A right triangle has legs 5 and 12. Find the hypotenuse.

Solution: $a^2 + b^2 = c^2$. $25 + 144 = 169$. $c = \sqrt{169} = \mathbf{13}$.

Example 4:

Find the area of a circle with diameter 10. ($\pi \approx 3.14$)

Solution: Radius = $10 \div 2 = 5$. Area = $\pi r^2 = 3.14 \times 25 = \mathbf{78.5 \text{ square units}}$.

Section 4: Practice Drill — 35 GED-Style Problems

Directions: Solve each problem. Show your work. You may use a calculator on problems 6–35 (just like the real GED).

Fractions & Percentages (1–12):

1. Simplify: $\frac{3}{4} + \frac{1}{6}$.
2. Simplify: $\frac{5}{8} - \frac{1}{4}$.
3. Multiply: $\frac{2}{3} \times \frac{5}{9}$.
4. Divide: $\frac{7}{8} \div \frac{1}{4}$.
5. What is 40% of 250?
6. A \$80 item is 25% off. What is the sale price?
7. Convert 0.625 to a fraction in lowest terms.
8. What is 12% of 350?
9. A student scored 72 out of 90. What percent is that?

10. Convert $\frac{3}{8}$ to a decimal.

11. A population grew from 500 to 650. What is the percent increase?

12. Add: $1\frac{3}{4} + 2\frac{1}{2}$.

Algebra (13–24):

13. Solve: $4x - 7 = 21$.

14. Solve: $2(x + 5) = 16$.

15. Solve: $x/5 + 3 = 8$.

16. Solve: $3x + 4 = x + 12$.

17. Solve: $5(x - 2) = 3x + 6$.

18. Solve: $2x + 9 = -1$.

19. Evaluate: $3x^2 - 2x + 1$ when $x = 2$.

20. Solve: $x/2 - 4 = 6$.

21. Solve: $6x - 3(x + 2) = 12$.

22. A number doubled, then increased by 5, equals 23. Find the number.

23. If $y = 2x - 3$, find y when $x = 7$.

24. Solve: $4(x + 3) = 2(x + 9)$.

Geometry & Mixed (25–35):

- 25.** Find the area of a rectangle: $l = 13$ m, $w = 7$ m.
- 26.** Find the perimeter of a square with side 11 cm.
- 27.** Find the area of a triangle: $b = 14$ in, $h = 6$ in.
- 28.** Find the circumference of a circle: $r = 4$ ft. ($\pi \approx 3.14$)
- 29.** Find the volume of a rectangular box: $l=6$, $w=4$, $h=3$.
- 30.** A right triangle has legs 9 and 12. Find the hypotenuse.
- 31.** Find the area of a circle: $r = 8$ m. ($\pi \approx 3.14$)
- 32.** What is the volume of a cylinder: $r=5$, $h=10$? ($\pi \approx 3.14$)
- 33.** A rectangle has perimeter 40 ft. Length is 12 ft. Find width.
- 34.** Find the hypotenuse: legs = 6 and 8.
- 35.** A garden is 25 ft by 18 ft. How many square feet of sod are needed?

Section 5: Answer Key

1. $9/12 + 2/12 = 11/12$
2. $5/8 - 2/8 = 3/8$
3. $10/27$
4. $7/8 \times 4/1 = 28/8 = 7/2 = 3 \frac{1}{2}$
5. $0.40 \times 250 = 100$
6. 25% of \$80 = \$20 off. Sale price = \$60
7. $5/8$
8. $0.12 \times 350 = 42$
9. $72/90 = 0.80 = 80\%$
10. 0.375
11. $(650-500)/500 \times 100 = 30\%$
12. $1 \frac{3}{4} + 2 \frac{2}{4} = 3 + 5/4 = 4 \frac{1}{4}$
13. $4x = 28 \rightarrow x = 7$
14. $2x + 10 = 16 \rightarrow 2x = 6 \rightarrow x = 3$
15. $x/5 = 5 \rightarrow x = 25$
16. $2x = 8 \rightarrow x = 4$
17. $5x - 10 = 3x + 6 \rightarrow 2x = 16 \rightarrow x = 8$
18. $2x = -10 \rightarrow x = -5$
19. $3(4) - 2(2) + 1 = 12 - 4 + 1 = 9$
20. $x/2 = 10 \rightarrow x = 20$
21. $6x - 3x - 6 = 12 \rightarrow 3x = 18 \rightarrow x = 6$
22. $2x + 5 = 23 \rightarrow x = 9$
23. $y = 2(7) - 3 = 11$
24. $4x + 12 = 2x + 18 \rightarrow 2x = 6 \rightarrow x = 3$
25. 91 m^2

26. 44 cm

27. 42 in²

28. $C = 2(3.14)(4) = 25.12$ ft

29. $V = 6 \times 4 \times 3 = 72$ cubic units

30. $c = \sqrt{(81+144)} = \sqrt{225} = 15$

31. $A = 3.14 \times 64 = 200.96$ m²

32. $V = 3.14 \times 25 \times 10 = 785$ cubic units

33. Width = $(40-24)/2 = 8$ ft

34. $c = \sqrt{(36+64)} = \sqrt{100} = 10$

35. $25 \times 18 = 450$ sq ft

